

Review

Recent Developments in Hydrogen Production, Storage, and Transportation: Challenges, Opportunities, and Perspectives

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Abstract: Hydrogen (H₂) is considered a suitable substitute for conventional energy sources because it is abundant and environmentally friendly. However, the widespread adoption of H₂ as an energy source poses several challenges in H₂ production, storage, safety, and transportation. Recent efforts to address these challenges have focused on improving the efficiency and cost-effectiveness of H₂ production methods, developing advanced storage technologies to ensure safe handling and transportation of H₂, and implementing comprehensive safety protocols. Furthermore, efforts are being made to integrate H₂ into the existing energy infrastructure and explore new opportunities for its application in various sectors such as transportation, industry, and residential applications. Overall, recent developments in H₂ production, storage, safety, and transportation have opened new avenues for the widespread adoption of H₂ as a clean and sustainable energy source. This review highlights potential solutions to overcome the challenges associated with H₂ production, storage, safety, and transportation. Additionally, it discusses opportunities to achieve a carbon-neutral society and reduce the dependence on fossil fuels.

Keywords: hydrogen production; biofuels; storage; safety; transportation; waste management



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1. Introduction

Large-scale use of renewable energy is a vital strategy for reducing fossil energy consumption and achieving ‘net zero’ emissions by 2050. Green hydrogen (H₂) is a promising technology that utilizes renewable energy sources [1]. H₂ may be produced through several processes, each with its own advantages and limitations. Standard methods of H₂ production include water (H₂O) electrolysis, steam methane (CH₄) reforming, biomass gasification, and biological processes [2,3]. The H₂O electrolysis process uses an electric current to split H₂O molecules into H₂ and oxygen (O₂) [4]. The steam CH₄ reforming process involves the heating of a natural gas and steam mixture to produce H₂ and carbon dioxide (CO₂) [5,6]. Biomass gasification converts organic materials such as agricultural waste or wood chips into a mixture of H₂, CO₂, and other gases through a thermochemical process [7,8]. Each method carries a different implication for carbon emission and sustainability. Additionally, emerging technologies such as photoelectrochemical electrolysis and algae-based biological H₂ production are promising alternatives for green H₂ production [9,10]. Thus, the development of efficient and sustainable methods for H₂ production by combining biological and non-biological procedures such as electrocatalytic and photocatalytic H₂O splitting is crucial for meeting the increasing global demand for energy and reducing reliance on fossil fuels [11,12].

Moreover, different types of H₂ production need to be considered based on the extraction methods [13]. Gray H₂ is produced through carbon-intensive processes such as coal or

natural gas gasification; blue H₂ is created by combining natural gas or coal gasification with carbon capture technologies to reduce carbon emissions; green H₂ is produced through H₂O electrolysis using renewable energy sources [13–15]. Gasification is one of the most prominent thermochemical technologies for converting biomass into renewable fuels and chemicals. It converts biomass into a mixture of gases, including H₂, carbon monoxide (CO), and CH₄, which may be used as energy sources [16,17]. H₂O electrolysis is one of the approaches to produce low-carbon H₂, which can generate lower emissions when compared to fossil-based or direct grid electrification [18]. Projections from the International Energy Agency indicate that 3.3 trillion worldwide installations for H₂O electrolysis technologies with a capacity of 15,000 TWh of electricity are required to achieve net zero emissions by 2050 [19]. The electrolysis process involves splitting H₂O into H₂ and O₂ with the help of electricity. Electrolyzers can be large-scale central production facilities directly connected to renewable energy systems or other non-greenhouse-gas-emitting forms of electricity production, or they can be small-sized devices ideal for small-scale hydrogen production [20]. The three types of electrolyzers commonly used for commercial scale operations are alkaline, proton exchange membrane (PEM), and solid oxide electrolyzer cell (SOEC). The significant difference between these three types of electrolyzers is that they are separators and electrolytes, with working temperatures of 700–800 °C for SOEC and 70–100 °C for PEM and alkaline fuel cells [19,21,22]. Based on production capacity and efficiency, commercial alkaline electrolyzer systems can have efficiencies between 43 and 69%. The cost of these systems can vary from 600 to 2600 €/kW, with most estimations being around 1100 €/kW [19]. Future alkaline electrolyzers are expected to require capital expenditures of 400–900 €/kW. Compared to PEM and SOEC, the primary benefits of alkaline electrolyzers are their low capital costs, huge plant sizes, and long life spans [19,23]. The cost of H₂ produced by electrolysis (green H₂) is comparatively high at \$4/kg over \$1.2/kg for fossil fuels (grey H₂) [24]. Soon, it can be expected that costs linked to electrolyzers and solar electricity will decline, and fossil fuel reforming costs associated with raw materials will be enhanced to favor electrolysis. Other important considerations include the transport and distribution of H₂, for which various H₂ production, storage, and application methods have been explored to meet the demands of an H₂-based economy [25,26]. H₂ storage is a crucial aspect of successfully implementing H₂ technology. The various H₂ storage techniques that have been explored include compressed gas storage, liquid H₂ storage, and solid-state storage [27,28]. Each storage method has different advantages and limitations in terms of efficiency, safety, and cost. Compressed gas storage involves compressing H₂ gas at high pressure into storage tanks [29,30]. Liquid H₂ storage involves liquifying H₂ by cooling it to extremely low temperatures for storage in cryogenic tanks [31,32]. Solid-state storage includes adsorptive storage on high-surface-area adsorbents, chemical storage in metals and complex hydrides, and storage in boranes [33,34]. Furthermore, H₂ storage technology has advanced to include reversible storage options and hydrolytic release of H₂ with off-board regeneration, which are solutions for potentially efficient and sustainable H₂ storage [35,36]. However, none of the existing storage methods fully satisfy the requirements of an H₂-based economy [37–39]. Therefore, ongoing research and development efforts are focused on improving the efficiency, safety, and cost-effectiveness of H₂ production and storage technologies [29,40]. This review discusses the detailed processes associated with H₂ production, storage, and transportation and presents the related recent updates, challenges, and perspectives.

2. Hydrogen Production

H₂ production is crucial for achieving a sustainable low-carbon future, and it requires the implementation of advanced methods and technologies [41]. Figure 1 illustrates the various H₂ production methods.



Figure 1. Illustration of various routes for H₂ production.

2.1. Thermochemical Routes

Thermal processes use energy from various resources such as biomass, coal, and natural gas to produce H₂. Alternatively, the heat associated with closed chemical cycles helps produce H₂ from H₂O (feedstock) [42]. Thermochemical processes such as biomass gasification, biomass-derived liquid reforming, natural gas reforming (steam CH₄ reforming; SMR), and solar thermochemical H₂ (STCH) have been used to produce H₂ [6,43]. The H₂ production by SMR process reached up to 74% in the industrial sector [44]. In recent years, biomass gasification has gained significant attention as an energy production process. In this process, biomass materials such as agricultural waste or wood chips are converted into a combustible gas mixture known as syngas (CO + H₂) [45]. Syngas is used as fuel for power generation or other industrial processes. Biomass gasification significantly reduces greenhouse gas (GHG) emissions compared with that of fossil fuel-based energy production [16,46]. Recently, the improvement in H₂ production through biomass gasification has been proposed via (i) minimizing tar/car formations using KCl as a catalyst, (ii) altering the temperature to enhance syngas composition, and (iii) enhancement in H₂ efficiency from up to 80% through capturing CO₂ using calcium oxide [44]. To achieve the zero-waste strategy, the thermally decomposed spent LiNi_xCo_yMn_{1-x-y}O₂ batteries-derived novel Ni/Co/Mn-loaded mesoporous Al₂O₃ catalyst exhibited significantly higher H₂ productivity 151% from a temperature of 600 to 800 °C using pinewood sawdust and polyethylene co-gasification over control catalysts (17.2 mmol/g) [47]. Additionally, developing advanced gasification technologies and integrating biomass gasification with other renewable energy sources such as solar or wind power may further enhance the efficiency and viability of biomass gasification [16,48]. Biomass gasification may also be used to manage waste by converting organic waste into valuable energy. Furthermore, integrating biomass gasification with carbon capture and storage technologies would potentially result in negative

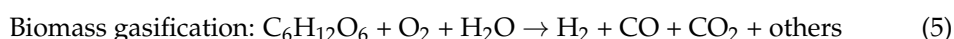
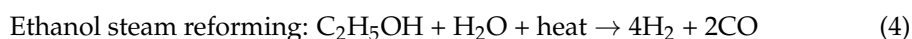
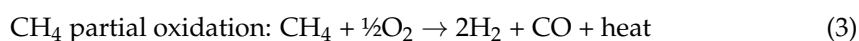
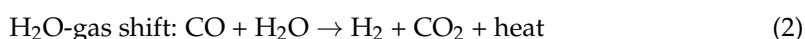
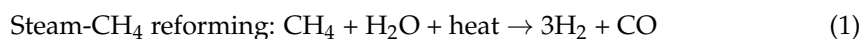
emissions. This makes gasification a useful tool for combating climate change [42]. The multi-source heat and power system, biomass gasification, solar collecting, SOEC, steam power, and gas turbine integrated systems achieved an energy efficiency of 64.5% and a feasible leveled cost of electricity of \$0.16/kWh [8].

The liquids obtained from biomass (ethanol, bio-oils, and other liquid biofuels) could be used to generate H_2 via a reforming process [13,49]. Compared with biomass, these liquids are easy to transport over long distances for H_2 production and distribution at fueling centers. This is a three-step process: (i) liquid fuels react with steam at high temperatures in the presence of catalysts to generate reformat gas (H_2 , CO, and CO_2); (ii) extra H_2 and CO_2 are generated from the reaction of CO and steam at high temperatures through an H_2O -gas shift reaction; and (iii) H_2 separation and purification [13,50]. The steam reforming of biomass-derived levulinic acid using 15Ni/ $NiAl_2O_4$ catalysts at 800 °C exhibited high carbon conversion, H_2 yield, and concentration of 96.3, 92.8, and 67.9%, respectively [51]. This type of process is a mid-term technology because large-scale biomass-based liquids can be produced at facilities near the sources (biomass feedstocks), which is economical as it reduces the cost of H_2 transportation to reforming sites [50]. However, the problematic reforming of larger molecules (more carbon atoms in liquid fuels than in natural gas (CH_4)) and the need for efficient catalysts to enhance selectivity and yield are a few major challenges associated with this process. Moreover, future studies are needed to identify ways to reduce the costs associated with biomass-based liquids, equipment, operation, and maintenance and enhance overall process efficiency [43,49,50].

Natural gas reforming gasification converts natural gas into synthetic gas [6]. This process is commonly used for producing H_2 , ammonia, and methanol. To produce 10 tons of liquid H_2 /day, the steam CH_4 reforming H_2 liquefaction and waste heat recovery system established on liquified cold natural gas showed specific energy consumption, coefficient of performance, and exergy efficiency of 5.93 kWh/kg of liquid H_2 , 0.22% and 53.24% under optimum conditions, respectively [52]. The challenges associated with the natural gas reforming gasification process include high energy consumption, the need for catalysts, and the management of CO_2 emissions [13,48]. Despite these challenges, natural gas reforming gasification has an excellent future perspective. Advancements in technology and increased focus on sustainable energy solutions have shaped natural gas reforming gasification into a potentially crucial means to reduce carbon emissions and transition to clean energy [6]. Integrating carbon sequestration schemes and using renewable energy sources in this process would further enhance its sustainability and contribution to the development of a circular economy [53]. Natural gas reforming is an advanced and mature production process that builds upon existing natural gas pipeline delivery infrastructure. Approximately 95% of the H_2 produced in the United States is produced by natural gas reforming in large central plants, making this an essential technological pathway for near-term H_2 production [6,53]. Chen et al. [54] evaluated the H_2 production cost estimation frameworks of SMR, surface coal gasification, and underground coal gasification with and without carbon capture. The H_2 production by SMR with 99% carbon capture is profitable, considering new production tax credits. Natural gas-based H_2 production through steam CH_4 reforming, autothermal reforming, and pyrolysis costs with and without carbon capture, utilization, and sequestration varied in the range of 1.48–2.55 and 1.22–2.12\$/kg of H_2 with production of 0.22–0.35 kg of H_2 /kg of fuel and a CO_2 -footprint of 1.84–11.4 kg of CO_2 /kg of H_2 , respectively [54].

Solar thermochemical processes use concentrated solar energy to generate high temperatures for use in thermochemical reactions to split H_2O and produce H_2 [55]. An enhancement in 30.8% of H_2 production at 600 °C by the solar-thermochemical process was reported compared to the thermochemical process (115 mmol/h/g) [56]. The STCH project was initiated in 2003 by the Department of Energy Office of Energy Efficiency and Renewable Energy (EERE, USA) to achieve the goal of H_2 production at \$2/kg [57]. This method offers several advantages, including the use of renewable solar energy and the potential for large-scale H_2 production. However, developing and implementing solar thermochem-

ical H₂ processes [39,58] involves challenges such as identifying suitable, efficient, and durable materials for reactors, improving conversion efficiency and reaction kinetics, and developing cost-effective systems that can be scaled up for industrial production [48,55]. Furthermore, integrating solar thermochemical H₂ processes into existing energy systems and infrastructures presents logistical challenges. Despite these challenges, the solar thermochemical H₂ process holds great promise for a sustainable clean-energy future [59,60]. Thus, the solar thermochemical process offers a viable pathway for large-scale H₂ production by harnessing concentrated solar energy. Finding solutions to the challenges associated with solar thermochemical H₂ processes through ongoing research and development could lead to improved efficiency, durability, and scalability of this technology [42]. Overall, the thermochemical H₂ production routes are a well-established and mature technology but still face limitations in generating GHGs and using non-renewable resources. Further, the advancement in these systems through the integration of carbon capture and storage technologies and the employment of renewable feedstock can be beneficial to achieve high H₂ production efficiency and a better environmental footprint. H₂ production via thermochemical reforming involves the following reactions (Equations (1)–(5)) [39,59,60]:



2.2. Electrolytic Routes

H₂ production through electrolytic routes has recently gained significant attention because of its potential as a clean and sustainable energy source [61]. The use of electricity to split H₂O into H₂ and O₂ through electrolysis provides a means to produce H₂ without producing greenhouse gases. Thus, this method could contribute to the reduction of carbon emissions and address the global challenges associated with reliable clean energy [4,62]. Furthermore, electrolysis provides flexibility in utilizing different electricity sources for the process, including renewable energy sources such as solar and wind power. Green H₂ thus produced can be used in various sectors including transportation, industry, and power generation [61]. Electrochemical H₂ production processes include electrolysis systems such as alkaline, PEM, SOEC, anion exchange membrane (AEM), bipolar membrane, and microbial electrolysis cells (MECs) (Figure 2) [23,62]. In electrochemical H₂ generation technologies, H₂O electrolysis is a crucial process that involves electric current passing through H₂O to convert into H₂ and O₂ in an electrolyzer system consisting of anode and cathode electrodes separated by an electrolyte. The H₂O is oxidized at the anode to produce O₂ and protons (H⁺) in the PEM, OH[−] in alkaline, or O₂[−] in SOEC processes. These produced ions migrate to the cathode through the electrolyte and are reduced to H₂ gas [23].

The alkaline electrolysis system is an extensively employed H₂O electrolysis technology to produce H₂. In principle, H₂ is produced in an alkaline electrolysis cell at the cathode through the electrochemical decomposition of H₂O operated by an electric current. In this system, adding alkali electrolytes such as sodium hydroxide or potassium hydroxide improved H₂ production efficiency by overcoming the limited intrinsic electrolytic properties of H₂O [23,62]. Asbestos materials are traditionally used to separate cathodes and anodes in the alkaline electrolysis system. Several different types of alkali electrolyzer systems are available, each with unique characteristics and advantages. PEM fuel cells and alkali electrolyzers are two types of electrochemical devices that play significant roles in producing and utilizing H₂ as a clean energy source [63,64]. These are crucial components of the H₂ economy and are used in various applications, such as fuel cell vehicles and stationary power generation. The proton-conducting membrane, as a solid polymer

electrolyte, is used in PEM electrolysis. It has higher energy efficiency and can operate at higher current densities than alkaline electrolysis [23]. They work together in a closed-loop system as a PEM/alkali electrolyzer system, where the PEM fuel cell utilizes H_2 to generate electricity and the alkali electrolyzer produces H_2 by splitting H_2O through electrolysis. This closed-loop system allows for the continuous production and utilization of H_2 , making it a sustainable and environmentally friendly energy option. Furthermore, this integrated system offers flexibility in adjusting the production and utilization of H_2 according to the specific needs and demands of the application. The forward osmosis- H_2O splitting system integrated with alkaline H_2O electrolysis showed high purity of H_2 with production at a rate of $448 \text{ Nm}^3/\text{day}/\text{m}^2$ from wastewater and specific energy consumption of $3.96 \text{ kWh}/\text{Nm}^3$ using 1 M of KOH [65]. Increasing wind-solar hybrid H_2 production by multi-state alkali electrolyzer management in 12 different scenarios, a daily average energy utilization rate reached 92.8%. This favors economic potential, with an internal return rate of 6.8% and an investment payback period of 12.9 years [66]. Aqueous room temperature synthesized transition metal dichalcogenide nanoparticles in a PEM H_2O electrolyzer accomplished $1 \text{ A}/\text{cm}^2$ of current densities at a voltage of 2.28 for MoS_2 and 2.07 for RuSe_2 with a long period of stability over 100 h [63]. These technologies provide a sustainable and efficient way to produce and utilize H_2 , contributing to the transition to a greener, more sustainable future. This H_2 production technology has been used for decades at various scales with relatively low production costs.

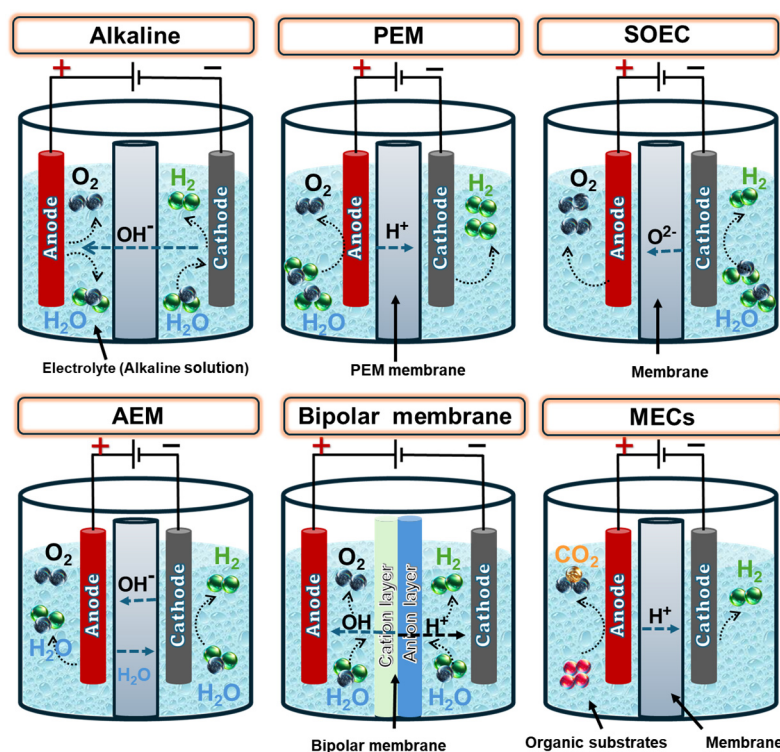


Figure 2. Illustration of H_2 production through various electrolytic routes.

Ceramic or SOEC electrolytes, i.e., yttria-stabilized zirconia, are used in SOEC that can be operational at high temperatures, up to 1000°C . In thermoneutral mode, the SOEC system integration, through industrial waste heat at a steam flow of $50 \text{ kg}/\text{h}$ and temperature of 800°C , resulted in maximum H_2 production of $2.47 \text{ kmol}/\text{h}$ [32,67]. Here, waste heat supply to total energy consumption percentage varied from 15.5 to 20.6%. This technology showed higher energy efficiency than other electrolysis processes, but it is a less mature commercial technology and requires a more complex system.

AEM, MECs, and bipolar membrane electrolysis are developed through the advancement of electrochemical technologies. The AEM system combines the advantages of PEM

and alkaline electrolysis that conduct OH^- as the electrolyte [21,23,64]. It uses non-precious metal catalysts at lower temperatures, making AEM more cost-effective than PEM. AEM-based alkaline seawater electrolysis using Co/P dual-doped bifunctional catalyst (Co/P- $\text{Fe}_3\text{O}_4\text{@IF}$) achieved a high current density of 500 mA/cm^2 at 1.83 V without chloride oxidation reaction, and a low energy of 4.38 KWh was spent to produce H_2/m^3 [68]. The inexpensive electrocatalyst bifunctional nickel copper phosphide-nickel electrocatalyst, as a non-platinum group metal, exhibited a performance of 30 mA/cm^2 at 2 V with excellent operational durability over 300 h in an AEM H_2O electrolyzer with a range of 95–97% retention rate and $80\text{--}800 \text{ mA/cm}^2$ current density [69].

The MEC system uses electroactive microbes to generate H_2 using organic feedstocks. In its mechanism, microbes oxidize organic substrates to create electrons and H^+ at the anode that flows towards the cathode through an external circuit and ion exchange membrane to form H_2 , respectively [21,23]. Polydopamine, polyethyleneimine, and silver nanoparticle-coated polyvinylidene fluoride membranes-based MECs exhibit antifouling properties and result in a high H_2 production rate of $2.65 \text{ m}^3/\text{m}^3/\text{d}$ by improving the proton transportation of membranes [70]. Pre-fermentation effluents-based MECs showed a 34.4% superior H_2 -producing potential of 911 mL/g of chemical O_2 demands (COD) than raw food wastewater [71]. Here, high H_2 production was associated with a 2-fold enhancement of *Geobacter* abundance at the anode compared to the control feed. These electrochemical processes offer various benefits, including (i) H_2 evolution without GHG emissions, (ii) excess of electricity from renewable sources, such as solar and wind, converts to H_2 , (iii) it can be scaled up depending on the diverse applications, and (iv) high-purity H_2 is produced, which can be directly used in fuel cells. Still, electrochemical technologies are under various levels to improve H_2 production yield, pilot-scale demonstration, and economy [21,23].

In bipolar membranes, electrolysis uses a composite membrane (anion/cation exchange layers) with a bipolar junction that allows easy H_2O dissociation and efficient H_2 evolution [23]. The bipolar membrane crosslinked with novel bipyridine covalent triazine frameworks exhibited maximum ionic conductivity of 143 and 48.0 mS/cm for H^+ and OH^- after dropping in $\text{H}_2\text{SO}_4/\text{NaOH}$ at 80°C , which resulted in a maximum current density of 1.29 A/cm^2 at 2.5 V [72]. The bipolar membrane consists of cation and anion exchange layers integrated into a bipolar membrane H_2O electrolyzer, which showed current densities of 250 mA/m^2 for splitting sea H_2O into H_2 and O_2 [73]. It inhibits the corrosive-free chloride generation by providing basic pH at the anode in contrast to PEM H_2O electrolyzers with acidic pH, which led to the failure of the PEM system in seawater. Overall, the electrochemical routes provide a clean H_2 , but enhancement in H_2 production efficiency and the cost-effectiveness of the system still require improvement at various levels. Ongoing development and research in these technologies, such as alkaline electrolysis and PME, are considered leading solutions to achieve economic and sustainable H_2 generation.

The electrolytic route of H_2 production offers several advantages, such as potentially low GHG emissions (because renewable energy sources power this process), ease of scaling up to meet the growing demand for H_2 in various sectors, and opportunity for energy storage by using the surplus electricity to produce more H_2 , which can be stored and used later during times of high energy demand or unavailability of renewable energy sources [59,61]. Nevertheless, recent developments in H_2 production via the electrolytic route need to be explored and the accompanying challenges and opportunities that arise in this field need to be understood for transitioning to a non-carbon-based economy [74,75]. Some significant progress achieved thus far includes: i) advancements in fabricating electrocatalytic materials, resulting in improved efficiency and cost-effectiveness, and ii) development of new synthesis methodologies, enabling the production of novel anodic and cathodic materials that are crucial for H_2 electrogeneration [59,74,76]. In summary, these advancements have paved the way for the widespread application of this technology to achieve an H_2 -based economy with the flexibility to use various sources of electricity, including renewable energy sources such as solar or wind power, to generate H_2 . Thus, electrolytic routes are essential for achieving a green and renewable H_2 production process [61,77].

2.3. Direct Solar Water-Splitting Routes

Photoelectrochemical (PEC) and photobiological processes use light as the energy source for splitting H_2O into H_2 and O_2 [5]. Currently, these processes offer long-term promise for sustainable H_2 generation with minimal environmental effects, and several are at the early stages of research [78]. PEC involves the use of specialized semiconductors, known as photoelectrochemical materials, and sunlight to split H_2O to generate H_2 . PEC semiconductor materials resemble photovoltaic systems (solar electricity generation), but these materials are dipped in H_2O -based electrolytes (sunlight energizes the H_2O -splitting) [59,79]. In the textile wastewater-integrated PEC system, a developed photoanode, $\text{WO}_3/\text{BiVO}_4/\text{Ni-PbS}$, showed a high current of 5.56 mA/cm^2 in phosphate buffer saline and 0.5 M of Na_2SO_3 at an external potential of 1.23 V with $52.2 \text{ } \mu\text{mol/cm}^2$ and simultaneous removal of 99.9% of total organic carbon [80]. In another study, MoSSe/MnSe heterostructure synthesized through a vacancy-controlled approach exhibited high PEC activity of -7.15 mA/cm^2 at 0 V with improved H_2 evolution reaction [81]. Liang et al. [82] demonstrated that the operating pressure influences bubble characteristics, bubble-induced optical losses, product gas crossover, and concentration overpotential, which is vital for PEC H_2O -splitting. Further, to achieve efficient PEC water-splitting, the pressure range can be 6–8 bar to minimize losses. The dual spin-controlled chiral two-/three-dimensional perovskite-based PEC H_2O -splitting system exhibited a high solar-to- H_2 efficiency of 12.5% [83]. The polymer carbon nitride ($\text{g-C}_3\text{N}_4$)-based photocatalyst $\text{R-TAP-Pd(II)@g-C}_3\text{N}_4$ PEC system showed a maximum H_2 generation rate of $1085 \text{ } \mu\text{mol/g/h}$ in visible light, which was 278-fold beneficial compared to the control catalyst as $\text{g-C}_3\text{N}_4$ [84]. In relation to the solar-to- H_2 pathway, PEC is a promising process that offers high conversion efficiency at low operational temperatures via particle semiconductor materials or cost-effective thin films, although PEC material cost, efficiency, and durability need to be improved to meet the desired market viability [78,85]. Ongoing research and development in this area are focused on: (i) H_2 production cost management by reducing the price of materials and processing, (ii) efficiency enhancement via improved absorption of sunlight and use of appropriate surface catalysis, and (iii) durability (lifetime) improvement through the use of a protective surface coat and desirable rugged materials [61]. In photobiological processes, microorganisms such as cyanobacteria and green microalgae convert H_2O (in some cases, organic matter) into H_2 [86] by splitting H_2O into O_2 and H_2 ions in the presence of sunlight. Then, the generated H_2 ions combine directly or indirectly to form H_2 molecules [87]. Bioelectrochemical systems (BES) use wastewater or biomass organic materials as substrates and electroactive microbes to produce H_2 in a bioreactor equipped with a cathode and anode separated by an ion exchange membrane [23]. In this mechanism, at the anode, the microbes oxidize substrates to generate electrons and H^+ that flow through an external circuit and an ion exchange membrane towards the cathode to form H_2 , respectively. A comparative analysis of bioelectrochemically and electric field-driven assisted dark fermentation reactors showed that extracellular electron transfer accounts for up to 76.3% of H_2 production in BES compared to 18.7% in the electric field-based systems [88]. At the pilot scale of 150-L, the nickel-foam-based cathode improved production to $19.1 \text{ L of H}_2/\text{m}^2/\text{d}$ under non-limiting substrate concentrations using industrial wastewater [89]. In BES, the H_2 production enhanced from 41.4 to 642 mM/mol of total organic carbon after treatment over control [90]. Overall, the advances in photocatalyst reaction efficiency and materials science can offer promising and clean H_2 production through photocatalytic routes. This emerging area can produce sustainable H_2 as a critical player based on future landscape considerations. The critical challenges of photobiological H_2 production include: (i) low production rate, (ii) released O_2 potentially inhibiting the H_2 production reaction and causing safety issues upon mixing with H_2 at specific concentrations, and (iii) low solar-to- H_2 efficiency [5,91,92]. To overcome these challenges, basic research has been focused on the following aspects: (i) selection of desirable strains to enhance H_2 production yield through efficient sunlight utilization, (ii) improvement of H_2 -producing enzymes

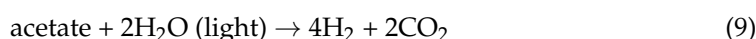
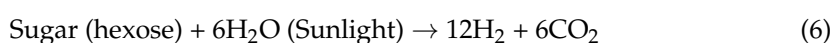
through modulation of metabolic pathways and protein engineering, and (iii) configuration and development of microbes and reactors for commercial H₂ production [79,93].

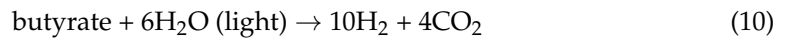
2.4. Biological Route

The biological route is a promising alternative to other existing methods of H₂ production [94]. These processes involve the microbial conversion of pure or biomass-derived sugars to H₂ through dark fermentation or photo-fermentation. Of these, dark fermentation shows better promise and sustainability for generating H₂ [95,96]. This process involves the anaerobic fermentation of organic substrates by dark fermentative bacteria, resulting in H₂ production as a byproduct [97]. Using sugars or biowaste feed, a maximum output of 3.8 mol of H₂/mol of hexose was obtained [95]. *Clostridium*, *Bacillus*, *Caldicellulosiruptor*, *Thermotoga*, *Citrobacter*, *Enterobacter*, and *Escherichia* have been studied as part of significant culture studies on dark fermentative H₂ production. The non-heat pretreatment sludge dominated by acetic acid-type fermentative *Thermoanaerobacterium* showed high H₂ production of 3.68 mol/mol of hexose without controlling pH under thermophilic conditions at 55 °C [98]. Recent developments in this field have focused on enhancing the efficiency and feasibility of dark fermentative H₂ production using various strategies such as optimizing reactor design and operating conditions, exploring novel substrates (co-digestion), improving microbial consortia, and genetically engineering bacteria [99–101]. These advancements have helped overcome challenges such as low H₂ yield and limited substrate availability. Thus, dark fermentative H₂ production is a viable option for sustainable energy production [102,103]. Moreover, integrating dark fermentative H₂ production with other bioprocesses such as wastewater treatment or organic waste management presents additional opportunities for maximizing overall energy output and resource utilization [104]. Nevertheless, challenges that need to be addressed for further development and commercialization of this technology include optimization of operating conditions and reactor design, enhancing microbial H₂ production efficiency, addressing substrate limitations, and improving process scalability [105,106]. Furthermore, integrating dark fermentative H₂ production with other renewable energy technologies such as biofuel and biogas (CH₄ + CO₂) production presents additional opportunities for creating sustainable and efficient energy systems [97,107,108]. Addressing these challenges and capitalizing on these opportunities would enable dark fermentative H₂ production to play a significant role in the transition toward a future with more sustainable carbon-neutral energy [106,109]. Recent developments in dark fermentative H₂ production have shown high potential for overcoming challenges and creating new opportunities for sustainable energy production [110,111]. The co-digestion of swine manure and food waste with tuning operational dark fermentation conditions resulted in a high purity of 83.2% and H₂ yields of 276 L/kg of volatile solids (VS) [112]. Designed microbial consortia of *Clostridium pasteurianum*, *Rhodopseudomonas palustris*, and *Syntrophomonas wolfei* showed ~4-fold higher H₂ yields of 78.6 mmol/g of COD than the maximum H₂ production by individual culture [113]. Bioprocess and metabolic engineering of *Clostridium tyrobutyricum* through endogenous hydrogenase overexpression resulted in the highest H₂ production of 910 mmol/L from molasses [114]. Dark fermentative H₂ production by *Enterobacter aerogenes* using immobilization supports such as ceramic balls, Persian Gulf dead coral, and ceramic saddles resulted in maximum H₂ production of up to 3.11 mol/mol of hexose [115]. Under a two-stage dark fermentation and AD process, the immobilized mixed culture showed high H₂ and CH₄ production of 21.3 and 374 L/g vs. at a high feed loading of 80 g of COD/L of food waste hydrolysate [116]. Ongoing research is aimed at further improving efficiency, increasing H₂ yield, and developing more cost-effective and scalable processes [102,117,118] to achieve a more sustainable future and transition to renewable energy sources.

Photo-fermentative H₂ production is a promising renewable-energy technology that produces H₂ [2,97] using photosynthetic bacteria or microalgae to convert organic compounds to H₂ in the presence of light. This technology may revolutionize the method of generating and storing clean energy [87,119] as it can harness the unlimited power of sunlight to generate a clean and sustainable source of hydrogen gas. Photofermentative

H₂ production has several advantages, namely, high energy conversion efficiency, low carbon footprint, and utilization of diverse feedstocks [120,121]. Under photo-fermentative conditions, *R. capsulatus* JP 91 produced a maximum of 9.0 mol of H₂/mol of glucose [122]. Important culture studies on photofermentative H₂ production included *Rhodobacter* spp. (*R. sphaeroides* and *R. capsulatus*) and *Rhodopseudomonas* spp. (*R. palustris* and *R. faecalis*). However, several challenges and opportunities for further development remain in this field. A significant challenge is the low efficiency of photo-fermentative H₂ production, owing to factors such as low solar energy to chemical energy conversion rates, limitations in the biological systems used, and the need for optimal growth and H₂ production conditions [121,123]. Various strategies have been attempted to overcome these challenges, such as genetic engineering of bacteria to enhance their H₂ production capabilities, optimizing growth conditions and nutrient availability, supplementing the production media with nanoparticles, and developing novel reactor designs to improve H₂ production efficiency [124,125]. Additional areas of focus for further development include scalability and cost-effectiveness of photo-fermentative H₂ production [120]. Furthermore, new strains or co-cultures of photosynthetic bacteria with higher H₂ production potential are being explored [126]. Addressing these challenges could potentially enable photo-fermentative H₂ production to become a sustainable and economically viable method for producing clean energy [2,127,128]. Punriboon et al. [129] showed a notable enhancement in H₂ production efficiency of 17-fold (493 mL/L) for *Rhodobacter pentothentaxigens* KKU-SN1/1 under outdoor conditions over conventional batch mode production. The growth and H₂ production potential of *Rhodobacter sphaeroides* KKU-PS1 is highly influenced by light intensity. Here, red light proved beneficial in high biomass accumulation and green light has a 2-fold enhancement in photo-H₂ productivity compared to the control white light [130]. Multi-dimension metabolic engineering of *R. sphaeroides* showed a 4-fold enhancement in photo-fermentative H₂ production to 17.6 L/L (3.0 mol/mol of acetate) over base strain [131]. This technology can be integrated with existing wastewater treatment systems to allow simultaneous H₂ production and wastewater purification; thus, photo-fermentative H₂ production is a promising renewable energy technology that can contribute to a cleaner and more sustainable future [120,132]. The complete conversion of sugars to H₂ would theoretically produce 12 mol of H₂ from one mol of hexose. Under dark fermentation conditions, up to 4 and 2 moles of H₂ can be generated via acetate and butyrate reaction intermediates, respectively. Furthermore, the photo-fermentative process can produce up to 8 moles of H₂ from 2 moles of acetate [133,134]. Thus, an integrative approach of dark fermentation followed by photo-fermentation of *Chlamydomonas reinhardtii* biomass yields a high H₂ output of 8.3 mol of H₂/mol of hexose [95,135]. The continuous sequential and co-culture dark-photo fermentations using synthetic lignocellulose hydrolysate demonstrated that the sequential process of dark and photo-fermentation exhibited a higher yield of 4.15 mol of H₂/mol of hexose than co-culture dark + photo fermentation with H₂ yield of 2.88 mol/mol [136]. In another study, combined dark and photo-fermentation of potato waste by *E. aerogenes* MTCC2822 and *R. sphaeroides* O.U.001 resulted in higher H₂ yields of 6.31 mol/mol of hexose [137]. The biological H₂-producing routes are at pilot-scale research and development with limited commercial production. The biological routes offer an eco-friendly approach to producing H₂ but still face limitations in production efficiency and bioprocess scalability. Biotechnological advancements in these routes will be unlocked to achieve efficient H₂ through a sustainable approach in the near future. Overall, H₂ production through dark and photo-fermentative routes involves the following reactions (Equations (6)–(10)) [95,133–135]:





The detailed benefits and disadvantages of various H₂ production routes are listed in Table 1.

Table 1. Hydrogen (H₂) production routes, reaction conditions, benefits, and disadvantages.

Process	Conditions	Benefits	Disadvantages	Reference
Biomass gasification	600–1000 °C	Highly efficient process for H ₂ production and operational at commercial scale	High capital and operational cost, catalyst deactivation, high risk of corrosion via slug formation, impure H ₂ production, high energy input, and CO ₂ emission	[8,46]
Ethanol steam-reforming	700–850 °C and 30–250 MPa	High-grade H ₂ production, mature production technology, and operational at commercial scale	High cost of metal catalysts, catalyst deactivation via coke deposition, high energy consumption, high purity required for feedstock, and high air/CO ₂ emission	[6,138]
CH ₄ partial oxidation	1150–1500 °C	Economically attractive process with no heat requirement, feasible at high CH ₄ concentration, no catalysts required, fast start-up/short response period, and entirely operational at commercial scale	Catalyst deactivation via coke deposition, low H ₂ /CO ratio, and CO ₂ emission	[6,30]
Autothermal-reforming	800–100 °C and 40 MPa	Low operation temperature, economically attractive process with no heat requirement, and feasible at CH ₄ concentration	Requires O ₂ /air, catalyst deactivation, limited commercial feasibility	[139,140]
Plasma-reforming	>2000 °C	High conversion efficiency and no catalysts required	The extensive energy input required to generate plasma, high electrode erosion, and CO ₂ emission	[141]
Supercritical H ₂ O gasification	350–600 °C and 22 MPa	High conversion efficiency, operation with biomass having high moisture, and low tar formation	High capital cost and energy input, costly feedstock harvesting, and CO ₂ emission	[142,143]
Solar thermochemical H ₂	600–1450 °C	High solar energy conversion efficiency	Solar system constrains due to high reaction temperatures, requires advanced materials to achieve high efficiency, and high operational and component costs	[57]
Alkaline electrolysis	60–90 °C and 20–100 MPa	Low capital cost, no catalysts requirement, and fully commercial phase technology with electrolytic efficiency of 60–75%	Extensive energy input for electrolysis operation at low temperatures, corrosive electrolyte environments, low current density generation, high maintenance cost, and long response time	[44,76]
Proton exchange membrane electrolysis	50–90 °C and 150–300 MPa	Simple design, early on the commercial phase production, and emerging technology with electrolytic efficiency of 70–90%	Extensive energy input for electrolysis operation at low temperatures, use of costly membranes and catalysts, operational in acidic environments, and immature technology	[4,44]
Solid oxide electrolysis	500–1000 °C and <300 MPa	Energy efficient, no catalyst requirement, and eco-friendly emerging technology with electrolytic efficiency of 85–100%	High capital cost and energy input, bulky system design, and immature technology	[8,44]
Dark fermentation	Low temperature (≤70 °C) and low pressure	Low operating cost and energy input, easy reactor design, pilot-scale production, and CO ₂ -neutral	Slow bioprocess and lower yield than theoretical production	[95,124,144]
Photo-fermentation	Low temperature and low pressure	Low operating cost, pilot-scale production, and CO ₂ -neutral	High capital cost for enzymatic production, slow bioprocess, and high energy input	[123,127,134]

3. Hydrogen Storage

H₂ storage is a crucial aspect of its utilization as an alternative energy source [145]. H₂ storage is carried out through various technologies in three states, including gaseous, liquid, and solid. Broadly, H₂ can be stored through (i) physical-based mode, which includes compressed gas, cold/cryo compressed, and liquid H₂ storages, and (ii) materials-based mode, which includes adsorbent (metal-organic framework, MOF-5), liquid organic H₂ carriers (LOHCs), interstitial hydride (LaNi₅H₆), complex hydride (NaAlH₄), and chemical H₂ (NH₃BH₃) [23,146–149]. H₂ storage as gas physically requires high-pressure tanks (350–700 bar) and a cryogenic temperature of −253 °C for liquid condition storage. It can also be stored on surfaces or within solids through adsorption or absorption [146,150].

The gaseous state of H₂ storage is categorized as compressed and underground storage. Compressed H₂ storage is conducted at high pressure in cylinders (individual or in cascade) or large vessels to reduce volume and enhance storage capacity [23,146]. On the other hand, underground storage implies H₂ storing in geological structures or subsurface formations such as aquifers, salt caverns, and depleted oil/gas reservoirs. Aquifers are underground H₂O-bearing formations, and H₂ can be stored in their confined and unconfined form of origin. Alternatively, salt caverns are underground formations created as solution-mined and mined caverns consisting of salts with poor geological availability. The depleted oil/gas reservoirs pose a potential option for H₂ storage because of their well-established containment assets and vast spaces [23,151].

Liquid-state H₂ storage is carried out through cooling and liquefying gaseous H₂ under extremely low temperatures, such as in cryogenic tanks [147,148]. The handling and requirements of specialized equipment set-up, liquid H₂ storage, and transportation are similar but differ in infrastructure, period, and primary objectives. LOHCs have gained attention as promising options for H₂ storage and transport. The unique chemistry and catalytic reactions of organic hydrides are well-suited for creating an H₂ delivery infrastructure, especially to realize a future H₂ energy-based society that widely uses renewable energy sources [152]. Moreover, LOHCs allow the use of existing infrastructure for fuel storage and transportation, reducing the need for costly and extensive infrastructure development. Several types of LOHCs have been investigated for potential applications in H₂ storage and release [152,153]. These include formic acid, methanol, ethanol, toluene, and various liquid organic compounds with functional groups such as amines, amides, and imides. This type of storage system offers numerous benefits over conventional storage methods, primarily because of its ability to efficiently store and transport H₂ in liquid organic substances [154,155].

LOHCs exhibit improved properties and performance for H₂ storage and release. However, the selection of appropriate LOHCs depends on several factors, including specific applications, cost considerations, and safety requirements [148,156]. Significant research and development have occurred in recent years in the field of LOHCs [149,157], leading to the feasibility of proving LOHC technology in commercial demonstrators for stationary energy storage systems. However, significant efforts in fundamental and applied research are still required for H₂ delivery to H₂ filling stations or direct LOHC fuel cell applications [39]. The H₂ storage capacity of LOHCs such as benzene/cyclohexane, toluene/methylcyclohexane, naphthalene/decalin, dibenzyltoluene/perhydro-dibenzyltoluene, bisphenyl/bicyclohexyl, diphenylmethane/dicyclohexylmethane, carbazole/dodecahydro-carbazole, and N-ethylcarbazole/dodecahydro-nethylcarbazole is in the range of 5.8–7.2 wt% or 47.4–65.4 kg m^{−3} (Table 2) [158–161]. However, numerous challenges remain in the development an efficient LOHCs technology. These include: (i) difficulty in reducing the dehydrogenation temperature and energy consumption, (ii) development of stable, efficient, and economic catalysts for hydrogenation/dehydrogenation, (iii) extremely high cost of purchasing LOHCs materials, and (iv) assessment of systems for technical, economic, and environmental performance [147,157]. Typically, petroleum- and coal-derived materials are used to prepare LOHC systems. Owing to the restrained quantum of traditional energy, novel approaches for manufacturing

procedures must be evaluated [159,162]. A promising approach would be the coupling of LOHCs with technologies related to the catalytic pyrolysis of biomass, given that benzene is the fundamental unit of biomass. Exploring orientation and research ideas is vital for the development of biomass-based LOHCs [163,164]. Appropriate technologies should be established to fabricate LOHC according to the specific reaction equipment used. The use of LOHCs could effectively address the energy storage issue in the transition to renewable energy. Additionally, LOHC applications extend beyond storage and transportation [157,165]. They may also be used for stationary energy storage, H₂ logistics, and on-board H₂ production in various mobile applications. Overall, LOHCs offer a promising solution for the storage and transportation of H₂, addressing the challenges associated with their low energy density and difficult handling. Further research and development in this area would be crucial for the widespread commercial success and implementation of LOHCs [39,159,166,167].

Table 2. Hydrogen storage materials and properties.

LOHC System (Storage/Carrier)	Storage Capacity (wt%)/Energy Density (kJ/g)	Number of H ₂ Stored	Reference
Benzene (C ₆ H ₆)/cyclohexane (C ₆ H ₁₂)	7.19/- ^a	3	[158,168]
Toluene (C ₇ H ₈)/methylcyclohexane (C ₇ H ₁₄)	6.20/7.39	3	[168]
2-Methylindole (C ₉ H ₉ N)/8H-2-methylindole (C ₉ H ₁₇ N)	5.76/-	4	[159]
Naphthalene (C ₁₀ H ₈)/decalin (C ₁₀ H ₁₈)	7.30/8.75	5	[35,169]
Biphenyl (C ₁₂ H ₁₀)/bicyclohexyl (C ₁₂ H ₂₂)	7.27/8.76	6	[29,156]
Carbazole (C ₁₂ H ₈ N)/dodecahydro-carbazole (C ₁₂ H ₂₀ N)	6.70/-	6	[159]
Diphenylmethane (C ₁₃ H ₁₂)/dicyclohexylmethane (C ₁₃ H ₂₄)	6.60/-	6	[29]
N-Propylcarbazole (C ₁₅ H ₁₅ N)/12H-N-propylcarbazole (C ₁₅ H ₂₇ N)	5.43/6.55	6	[159]
Perhydro-N-ethylcarbazole (C ₁₄ H ₁₃ N)/N-ethylcarbazole (C ₁₄ H ₂₅ N)	5.80/7.00	6	[168]
<i>cis</i> -Perhydro-1-(N-phenylethyl)naphthalene/ <i>trans</i> -Perhydro-1-(N-phenylethyl)naphthalene	6.49/-	8	[153]
Dibenzyltoluene (C ₂₁ H ₂₀)/perhydrodibenzyltoluene (C ₂₁ H ₃₈)	6.20/7.50	9	[36,170]

^a Not available or reported.

Lignocellulosic biomass contains lignin (15–30%), which can be easily depolymerized into aromatic compound intermediates via a thermochemical conversion pathway [163,171]. Therefore, these aromatic compounds can potentially be used for storing and releasing H₂ via a pair of reversible reactions using LOHC technology [148,159]. Biomass thermal conversion conditions, such as targeted deconstruction, nitrogen doping, better catalysts, and reaction conditions, must be optimized for LOHCs. Recently, a high-silica HY zeolite-supported RuW alloy catalyst enabled the conversion of lignin to benzene with an 18.8% yield [163]. Recent significant developments in H₂ storage have occurred in relation to compressed H₂ storage tanks, liquid H₂ storage tanks, and hydride containment systems, leading to potential advantages such as increased storage capacity, improved safety measures, and enhanced efficiency in storing H₂ [157,159]. Despite these improvements, challenges and opportunities persist [39,172]. A primary challenge in H₂ storage is identifying materials that enable the safe and compact storage of H₂. In fact, the search for lightweight materials that can safely store H₂ remains a bottleneck in the H₂ economy [173]. Additionally, the cost of storage systems, particularly those capable of maintaining high-pressure H₂, is

concerning. Despite these challenges, advancements in H_2 storage, such as the discovery of new materials, namely, complex hydrides and boranes, which offer reversible storage options and efficient H_2 release, have been reported [169,174]. Furthermore, innovative technologies, such as nanoparticles and 3D-supported nanomaterials, hold promise for improving H_2 storage capabilities and enhancing the feasibility of accessing underground storage [42,155,175]. Compared to gaseous H_2 , liquid H_2 has a higher volumetric density, making transportation and storage easier. H_2 carriers-based transportation offers lower flammability or explosion risk than liquid or gaseous forms of H_2 , which enhances safety during storage and transportation [23,151]. Developing and advancing liquefaction technologies can reduce energy inputs for converting gaseous H_2 into its liquid form. This technology can be integrated through renewable energy systems and its future exploration for suitability in the space industry.

The solid-state H_2 storage uses solid materials or compounds such as chemical and metal hydrides or H_2 absorbed on/within porous materials [23]. In metal hydride storage methods, H_2 directly interacts with metals or their alloys, such as AB_2 or AB_5 types, to form reversible and stable bonds that allow easy release of H_2 through decreasing pressure or heating. In contrast, complex hydrides consist of metals or metalloid elements attached to H_2 , such as boron [i.e., $LiBH_4$, $NaBH_4$, and $(Li_2Mg(BH_4)_4)$], aluminum (i.e., $NaAlH_4$ and $LiAlH_4$), and nitrogen [i.e., $LiNH_2$, $LiNH_2$, NH_3BH_3 , and $Mg(NH_2)_2$] [23,176]. The adsorbent materials such as zeolites, metal-organic frameworks, and activated carbons stored H_2 through physisorption. This technology operates at low temperatures and pressures over liquid or gaseous H_2 storage methods by achieving high density and minimal infrastructure requirements [23]. Figure 3 illustrates the various H_2 storage-associated processes.

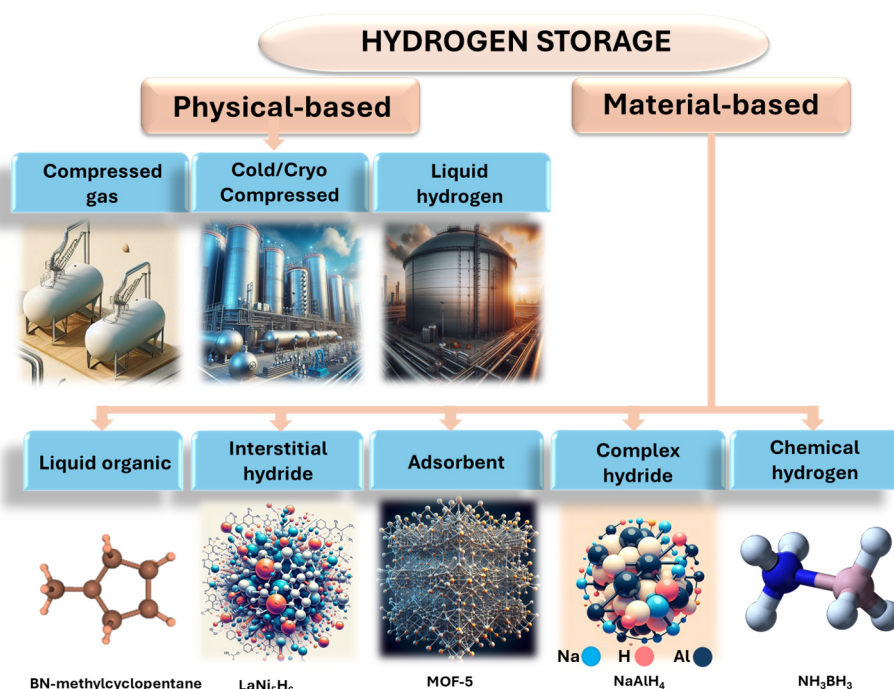


Figure 3. Illustration of H_2 storage processes.

4. Hydrogen Transportation

H_2 transportation has gained significant attention in recent years as a potential solution for reducing carbon emissions and addressing environmental concerns. With growing interest in renewable energy and the need for cleaner transportation options, H_2 has emerged as a promising alternative fuel. Recent developments in H_2 transportation have led to promising advancements in the use of H_2 as a fuel for vehicles [25,177,178]. H_2 transport involves various techniques for moving H_2 from its point of generation to the point of use. H_2 transportation is carried out through tank truckers, ships, pipelines, and

tube trailers. Transportation of H_2 is a crucial aspect for consideration in the economy. Various H_2 transportation methods are employed, including (i) gaseous H_2 transportation, which is used in the gas state of H_2 transport, (ii) liquid H_2 transportation, which requires cooling and liquefaction of H_2 at very low temperatures, and (iii) carrier H_2 transportation, which uses compounds or materials to capture and release H_2 [23,150,179].

In the gaseous method, H_2 is compressed using high pressure for transportation in well-designed containers with safety standards [151]. The transport of H_2 as a gas is suitable for low-volume applications because of the low H_2 density in atmospheric pressure. This method is widely employed for H_2 transportation through pipelines and high-pressure tube trailers due to its low cost and well-developed infrastructure. Gas compression methods commonly use mechanical (piston- and diaphragm-compressors) and centrifugal compressions [23,180]. Ionic liquid pistons and electrochemical compressions are advanced technology. The transport of H_2 through pipelines can be a high capital cost due to their construction and periodic maintenance requirements, including safety measurements [23,25,176].

Liquid H_2 transportation involves the transport of H_2 in a fluid state using cryogenic tankers that convert gaseous H_2 to a liquid form through cooling at an extremely low temperature of $-253\text{ }^\circ\text{C}$ [23]. Ships equipped with cryogenic tanks and insulation systems are also being created for the long-distance transport of fluid H_2 in bulk. Traditional processes of H_2 liquefaction require significant energy inputs and are carried out in several steps of cooling and compression systems. The various H_2 liquefaction technologies include magnetic refrigeration, thermoacoustic, Linde-Hapson, Claude, Brayton, and two-stage mixed refrigerant cycles [23]. Besides gaseous and liquid H_2 transportation, H_2 carrier transportation offers an alternative method, using materials or compounds such as ammonia, formic acid, $Mg(BH_4)_2$, metal hydrides, and LOHCs. This type of technology is at the developmental stage over high-pressure tube trailers and pipelines such as matured or liquid transportation as an advanced technology [23,181]. H_2 reacts with other elements to form various compounds, which allows it to be transported in its solid form [30,182]. For example, the transport of atomic silicon (produced using renewable energy) enables the separation of H_2O into H_2 and O_2 . Subsequently, H_2 is oxidized with O_2 (or air) to produce energy, with H_2O as the only byproduct. Mechanochemistry refers to chemical reactions that are triggered by mechanical forces rather than heat, light, or electric potentials [177,178]. Techniques such as ball milling crush materials (e.g., boron nitride or graphene), allowing them to absorb H_2 gas. The stored H_2 is released by heating the powder. These methods have the potential for substantial net energy savings. However, several challenges and opportunities that are associated with H_2 transport must be addressed [39,172]. The infrastructure required to support H_2 transportation is a major challenge. This includes the need for H_2 production facilities, storage tanks, and a network of refueling stations. Additionally, the cost of producing and distributing H_2 continues to stay relatively high compared with that of traditional fuels. Nevertheless, opportunities for innovation and collaboration include the development of cost-effective production methods and the expansion of the infrastructure network [101,178]. Moreover, H_2 transportation faces technical challenges regarding storage and safety. Thus, developing efficient and safe H_2 storage systems is crucial for its widespread adoption in the field of transportation. H_2 transportation offers the potential for zero-emission vehicles and reduces dependency on fossil fuels [13,183]. Furthermore, H_2 provides longer-range capabilities for vehicles and heavier-duty trucks, which addresses the limitations of battery-electric vehicles in terms of range and load capacity. Another opportunity for H_2 transportation is its potential for use as a chemical feedstock in various industries. Thus, H_2 may be used as a clean and versatile input for chemical processes, enabling the decarbonization of industries such as steel production and fertilizer manufacturing [30,178]. Although challenges such as infrastructure development and storage solutions need to be overcome, significant opportunities are available for H_2 transportation. These opportunities include reducing carbon emissions, achieving zero-emission vehicles, and providing longer-range capabilities for transportation [39,184]. Ongoing development and research can aim for

emerging methods to enhance efficiency, cost-effectiveness, and safety in H_2 transportation technology [23]. Figure 4 illustrates the various processes involved in H_2 transportation.

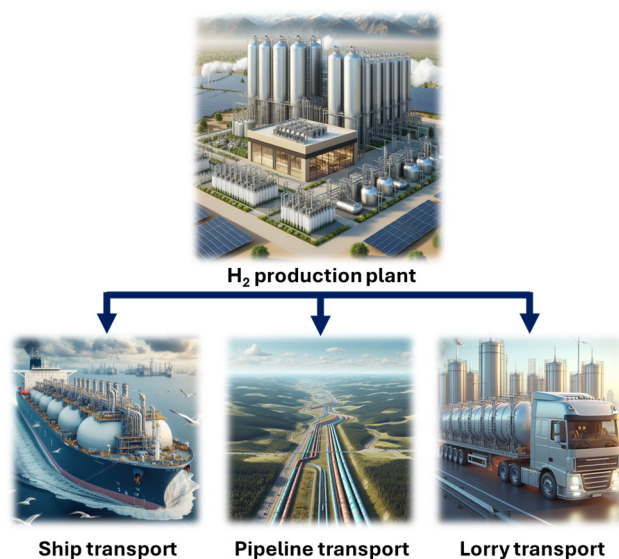


Figure 4. Illustration of H_2 transportation processes.

5. Challenges, Opportunities, and Future Perspectives

The development of sustainable energy technologies, such as green approaches to producing biofuels, is required to address energy, resource, and environmental issues [185–188]. Various approaches have been demonstrated to use waste to develop biofuels and valuable chemicals in a circular economy [108,171,189,190]. H_2 production is a key area of focus in the quest for sustainable and clean energy sources [97,191]. Different methods are available for producing H_2 , and one promising approach is the solar thermochemical process [55,59]. Recent developments in H_2 production via thermochemical routes have opened new possibilities and challenges in the field of H_2 economy by presenting alternative methods for large-scale H_2 production using various thermochemical processes such as H_2O -splitting, CH_4 -reforming, and biomass gasification [39]. These thermochemical routes provide sustainable and efficient H_2 production opportunities while using renewable resources and minimizing carbon emissions [43,58]. Moreover, these thermochemical routes can overcome some of the limitations associated with traditional H_2 production methods, such as high energy consumption and reliance on fossil fuels. However, the implementation of thermochemical routes for H_2 production is not without its challenges [17,48], such as the need to optimize reaction kinetics, improve process efficiency, and address the high temperatures and pressures needed for certain thermochemical reactions [53]. Integrating thermochemical H_2 production with the existing infrastructure and energy systems poses further challenges. Recent developments in thermochemical routes for H_2 production present opportunities and challenges that hinder the pursuit of a sustainable and efficient H_2 economy [53,58]. Through the use of biomass as a fuel source, the gasification process offers the advantages of using renewable and abundant resources. However, despite its potential benefits, biomass gasification faces several challenges that must be addressed before widespread adoption [7,8,46]. These challenges include the variability of biomass feedstock, the complex nature of gasification reactions, and the need for efficient syngas cleaning and conditioning. Therefore, further studies are needed to address these challenges and explore the full potential of biomass gasification to establish it as a viable and environment-friendly energy generation technique [7,58]. The development of efficient and sustainable methods for H_2 production, such as electrocatalytic and photocatalytic H_2O -splitting, is crucial for meeting the increasing global energy demand, reducing reliance on fossil fuels, and mitigating environmental issues [30,144]. The conversion of algal biomass into various

biofuel products is a promising strategy for sustainable energy solutions [192,193]. Dark fermentative H_2 production has emerged as a promising and sustainable method for generating H_2 . The use of dark fermentative bacteria for anaerobic fermentation of organic substrates has shown significant promise for producing H_2 as a clean and renewable energy source [87,194,195]. Recent developments in dark and photo-fermentative H_2 production have paved the way for a more efficient and sustainable method for generating H_2 [196]. These advancements have opened up new possibilities for integrating dark fermentative H_2 production with other renewable energy technologies, such as solar or wind power, to create integrated energy systems that are both reliable and sustainable. Additionally, the high feasibility of a circular economy has been assessed by developing biomass-based approaches to produce biofuels, including H_2 and value-added products [197–201]. However, the primary challenge in H_2 production processes is cost. Therefore, the EERE agency is focusing on developing methods to produce H_2 at \$2/kg and \$1/kg via net-zero-carbon pathways by 2026 and 2031, respectively [202,203]. All biological H_2 -producing routes are at the research and development pilot scale with limited commercial production. The recent advancement in wind/solar/ H_2O -electrolysis routes of H_2 production and technological upgradation can provide more economically green H_2 below \$2/kg than the biorefinery route [44,54,204]. With regard to H_2 storage technologies, some studies have explored the use of LOHCs such as formic acid, methanol, ethanol, toluene, and various types of liquid organic compounds with functional groups such as amines, amides, and imides [170,205,206]. LOHCs offer various advantages and disadvantages in terms of H_2 storage capacity, stability, ease of transport, and compatibility with the existing infrastructure [159,168,207]. Thus, developing effective and efficient H_2 storage methods is crucial for the widespread adoption of H_2 as an energy carrier [155]. The search for lightweight materials that enable safe and compact storage of H_2 continues to be a challenge in H_2 production [149,159,184]. In addition, developing an H_2 economy has garnered significant attention in recent years as a potential solution to address the pressing challenges of climate change and energy security [204]. Public awareness and acceptance of H_2 technologies are critical factors in successfully implementing an H_2 economy. Therefore, the transition toward an H_2 economy involves a complex interplay of technological, economic, and social factors [208,209]. Addressing safety concerns, demonstrating the environmental benefits, and fostering a positive public perception through education and outreach campaigns can help drive the widespread adoption of H_2 -based solutions [204,209]. Natural H_2 can be produced through various activities such as decomposition, bioactivity, serpentinization, human activity, degassing, H_2O reduction, radiolysis, and others [24]. Based on an estimate of peridotites in the Earth's upper crust, a cumulative 10^8 Mt of H_2 can be generated to supply 1000 Mt/year for 100,000 years. Still, the industrial use of natural H_2 is in its primitive stages. Therefore, the role of natural H_2 in energy transition may change the economic perspective significantly [24]. Overall, the major challenges in H_2 production may be overcome through infrastructure development, technological advancement, and cost-competitive strategies.

6. Conclusions

H_2 has emerged as a promising and sustainable alternative energy source with wide-ranging applications in various sectors such as transport and industry. Significant advancements have been achieved in H_2 production technologies by focusing on efficient and environment-friendly methods. These developments include the production of H_2 using renewable energy sources such as solar and wind power to perform electrolysis. Other recent developments include the implementation of novel catalysts and materials for enhanced H_2 production via various pathways such as H_2O -splitting, biomass conversion, and microbial processes. These advancements in H_2 production contribute to reducing GHG emissions and pave the way for a future that is more sustainable and carbon-neutral. The storage of H_2 is a complex and multidisciplinary field, and various methods and

technologies are being explored. Research and development are ongoing in the field of H₂ storage to improve its efficiency, safety, and cost-effectiveness.

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